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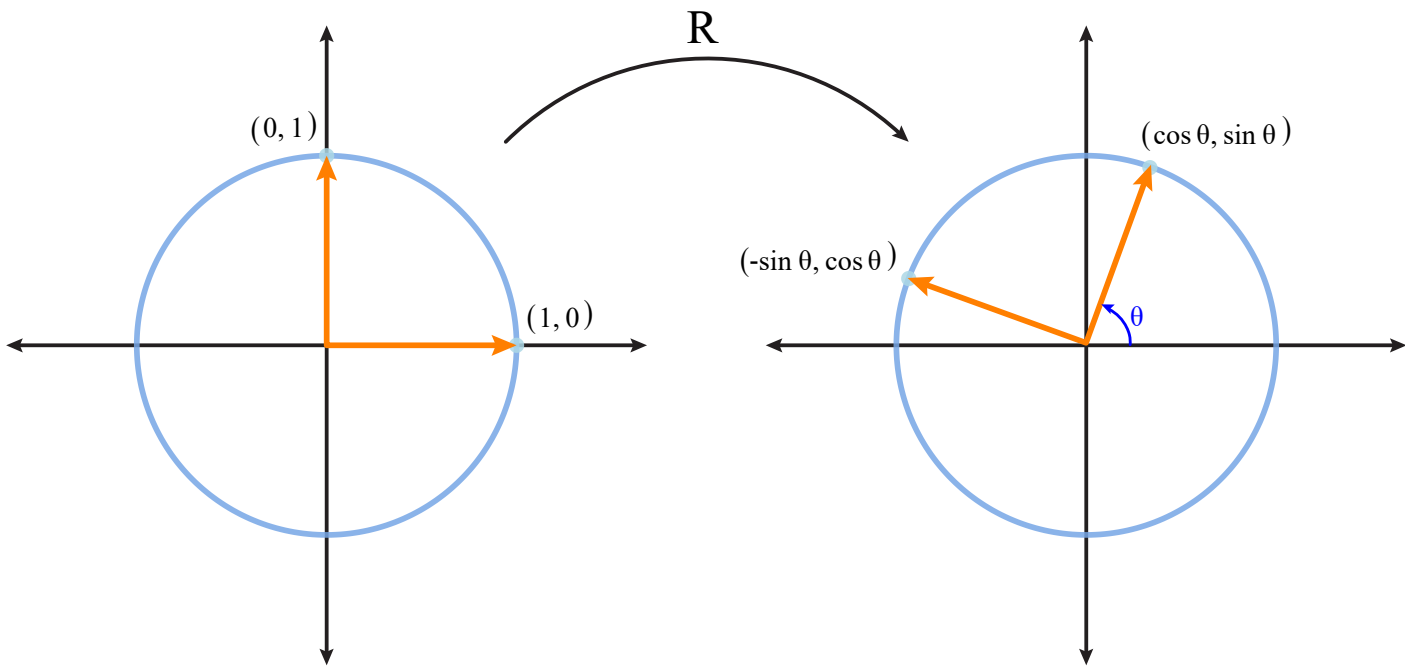
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Lecture due Jan 24, 2024 20:30 IST

It turns out that if we have a function from $\mathbb{R}^n$ to $\mathbb{R}^m$ that we know comes from multiplication on the left with a matrix $\mathbf{A}$, we can find this matrix by looking at where this matrix sends the standard basis vectors!

Problem 7.1 Given θ , there is a 2×2 matrix $\mathbf{R}$ whose associated function rotates each vector in $\mathbb{R}^2$ counterclockwise by the angle θ . What is it?



Solution: The rotation maps $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ to $\begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to $\begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$. We know that the matrix with columns $\mathbf{R} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{R} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is the matrix we want, and that this approach for finding a matrix representing a function always works. Thus

(first column of $\mathbf{R}$) = $\mathbf{R} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$

(second column of $\mathbf{R}$) = $\mathbf{R} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix},$

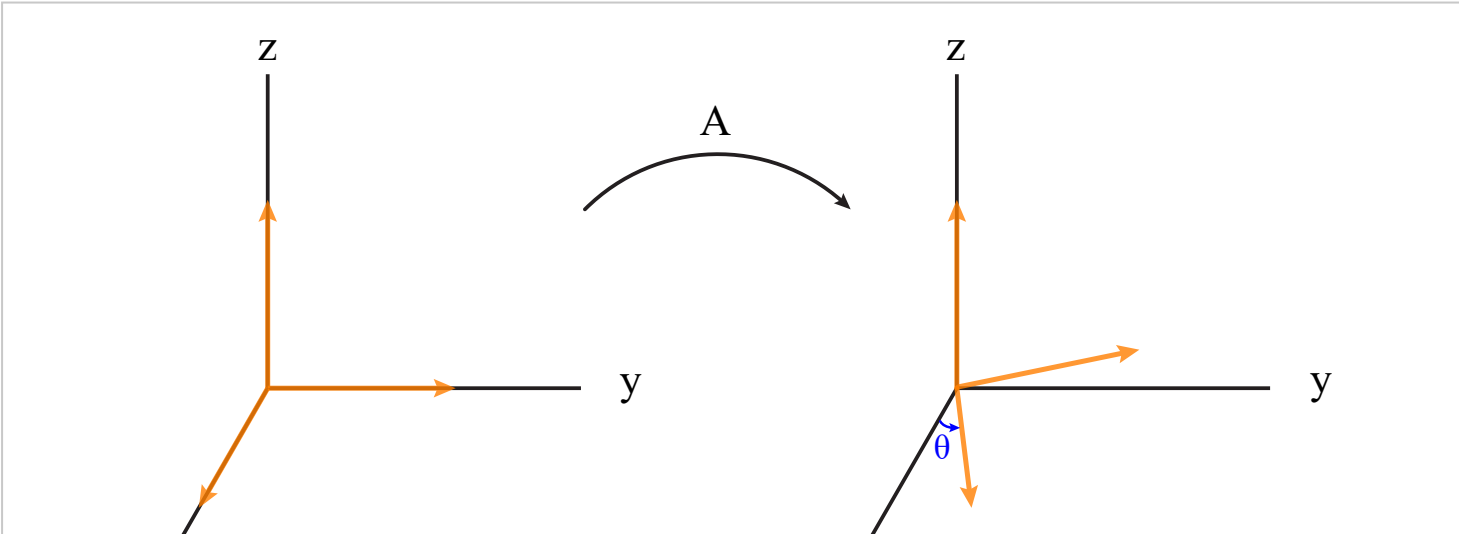
so

$$\mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Rotation in three dimensions

1/1 point (graded)

What is the matrix $\mathbf{A}$ which rotates the xy -plane in $\mathbb{R}^3$ counterclockwise by an angle θ about the z -axis?



/

X

/

X

☐

$$\begin{pmatrix} \cos \theta & 1 & 0 \\ \sin \theta & 0 & -\sin \theta \\ 0 & 0 & \cos \theta \end{pmatrix}.$$

☐

$$\begin{pmatrix} \cos \theta & -\sin \theta & \cos \theta \\ \sin \theta & \cos \theta & \sin \theta \\ \cos \theta & -\sin \theta & \cos \theta \end{pmatrix}.$$

☒

$$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

☐

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}.$$



Solution:

The z -axis is fixed by this rotation, so

$$\mathbf{A} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

On the other hand, the x and y basis vectors rotate by θ in the plane, but still don't have a z -coordinate after the rotation; only their x and y components change. So, similar to the example above,

$$\mathbf{A} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix},$$

and

$$\mathbf{A} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\sin \theta \\ \cos \theta \\ 0 \end{pmatrix},$$

so

$$\mathbf{A} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

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Answers are displayed within the problem

Depicting functions as matrices

1/1 point (graded)

The function associated to a matrix $\mathbf{A}$ sends the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ to the vector $\begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$, and sends

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the vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to the vector $\begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$.

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What is the first column of the matrix **A**?

☐ $\begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$

☒ $\begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$

☐ It cannot be determined without more information.

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☐ $\begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$

☒ $\begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$

☐ It cannot be determined without more information.



Solution:

The vector resulting from $\mathbf{A} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is actually the first column of **A**, so we get that the first column of **A** is $\mathbf{f} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$.

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Geometric examples in $\mathbb{R}^3$: Here are some functions from $\mathbb{R}^3$ to $\mathbb{R}^3$ that can be represented as a matrix: reflections across a plane through the origin, rotations about a line through the origin, and projections onto a plane or line through the origin.

Nonexample: A function from $\mathbb{R}^3$ to $\mathbb{R}^3$ which translates all points by a fixed amount cannot be represented by multiplication by a matrix.

(Optional) When can a function be represented by a matrix?

A matrix represents a very special kind of function called a **linear transformation**.

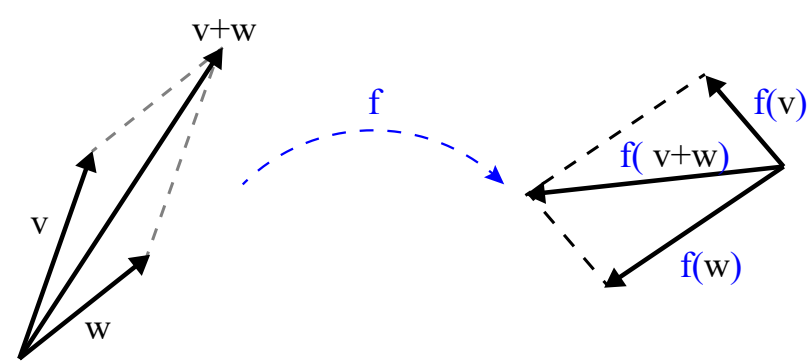
A function $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a **linear transformation** if:

- 0. it maps the zero vector in $\mathbb{R}^n$ to the zero vector in $\mathbb{R}^m$: $\mathbf{f}(\mathbf{0}) = \mathbf{0}$,
- 1. for a vector $\mathbf{v}$ and a constant c , $\mathbf{f}(c\mathbf{v}) = c\mathbf{f}(\mathbf{v})$, and
- 2. for vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$, $\mathbf{f}(\mathbf{v} + \mathbf{w}) = \mathbf{f}(\mathbf{v}) + \mathbf{f}(\mathbf{w})$.

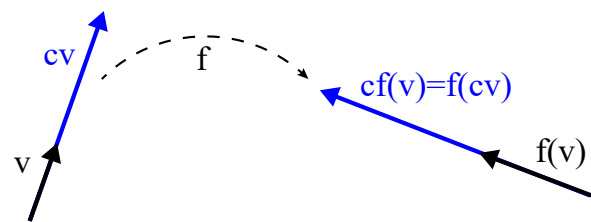
What this means roughly is that the function preserves vector addition and multiplication by a

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scalar. In pictures, we could draw this the following way.



If you take two vectors and add them, then take the linear transformation, this is the same as first taking the linear transformation of each of the two vectors, and then adding them.



If you take a vector, scale it, and then take its linear transformation, this is the same as first taking the linear transformation of the vector, and then scaling it.

Any function that is a linear transformation can be represented by a matrix by the method described above in the text.

To learn more about linear transformations, you may want to watch the following video.

[Linear transformations and matrices](#)

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7. Representing functions with matrices

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